

## Tasks on Jenkins-03

**1. Create one Jenkins job using the below code and create three stages.**

- **stage1: Git clone to download the source code.**
- **stage2: Sonarqube Integration to check the quality of code.**
- **stage3: Slack Integration to send the alerts to slack.**

**Source Code:** <https://github.com/betawins/VProfile-1.git>

### Implementation – Jenkins Freestyle Job with GitHub, SonarQube and Slack

1. Login to the Jenkins EC2 server and switch to the root user. `sudo su -i`
2. Verify Jenkins service and configured port. `systemctl status jenkins`
  - a. `ss -lntp | grep 8081`
3. Access Jenkins using the configured URL.
  - a. `http://<JENKINS-PUBLIC-IP>:8081`
4. Login to Jenkins and install the required plugins from Manage Jenkins → Plugins:
  - a. Git
  - b. SonarQube Scanner
  - c. Slack Notification
5. Login to the SonarQube EC2 server and verify the installed Java version.
  - a. `java -version`
6. Install Java 8 required for the selected SonarQube version.
  - a. `dnf install -y java-1.8.0-amazon-corretto`
7. Verify Java 8 installation.
  - a. `java -version`
8. Install Docker on the SonarQube server.
  - a. `dnf install -y docker`
  - b. `systemctl start docker`
  - c. `systemctl enable docker`
9. Pull the MySQL 5.7 Docker image.
  - a. `docker pull mysql:5.7`

10. Create and start the MySQL 5.7 container with the SonarQube database and user.

- a. `docker run -d --name mysql57 \`
- b. `-e MYSQL_ROOT_PASSWORD=Root@123 \`
- c. `-e MYSQL_DATABASE=sonar \`
- d. `-e MYSQL_USER=sonar \`
- e. `-e MYSQL_PASSWORD=Sonar@123 \`
- f. `-p 3306:3306 mysql:5.7`

11. Verify that the MySQL container is running.

- a. `docker ps`

12. Login to MySQL and verify the sonar database and sonar user.

- a. `docker exec -it mysql57 mysql -u root -p`
- b. `SHOW DATABASES;`
- c. `SELECT user,host FROM mysql.user;`

13. Download the required SonarQube ZIP package and copy/extract it under /opt.

- a. `cd /opt`
- b. `unzip sonarqube-*.zip`

14. Rename the extracted SonarQube directory.

- a. `mv sonarqube-* sonar`

15. Change the ownership of the SonarQube directory to ec2-user.

- a. `chown -R ec2-user:ec2-user /opt/sonar`

16. Open the SonarQube configuration file.

- a. `vi /opt/sonar/conf/sonar.properties`

17. Configure the SonarQube database username.

- a. `sonar.jdbc.username=sonar`

18. Configure the SonarQube database password.

- a. `sonar.jdbc.password=Sonar@123`

19. Configure the MySQL database connection.

- a. `sonar.jdbc.url=jdbc:mysql://127.0.0.1:3306/sonar`

20. Configure SonarQube to run on port 9000.

- a. `sonar.web.port=9000`

21. Switch to ec2-user and start SonarQube.

- a. `su - ec2-user`

- b. `cd /opt/sonar/bin/linux-x86-64`
  - c. `./sonar.sh start`
- 22. Verify SonarQube status.
  - a. `./sonar.sh status`
- 23. Verify SonarQube processes.
  - a. `ps -ef | grep -E 'sonar|elasticsearch|java' | grep -v grep`
- 24. Check SonarQube logs if required.
  - a. `tail -100 /opt/sonar/logs/sonar.log`
  - b. `tail -100 /opt/sonar/logs/es.log`
  - c. `tail -100 /opt/sonar/logs/web.log`
- 25. Access SonarQube using:
  - a. `http://<SONARQUBE-PUBLIC-IP>:9000/sonar`
- 26. Login to SonarQube and create an authentication token from My Account → Security → Tokens.
- 27. Login to Jenkins and configure the SonarQube server under Manage Jenkins → System.
- 28. Add the SonarQube server with the following details:
  - a. Name: My-SonarQube
  - b. Server URL: `http://<SONARQUBE-PUBLIC-IP>:9000/sonar`
  - c. Credential: sonarqube-token
- 29. Create the SonarQube token in Jenkins as a Secret Text credential.
  - a. ID: sonarqube-token
  - b. Secret: <SONARQUBE-TOKEN>
- 30. Configure SonarQube Scanner under Manage Jenkins → Tools → SonarQube Scanner installations.
- 31. Configure the SonarQube Scanner properties with the SonarQube server URL.
  - a. `sonar.host.url=http://<SONARQUBE-PUBLIC-IP>:9000/sonar`
- 32. Create a Slack channel named:
  - a. #Jenkins-bot
- 33. Configure Slack integration under Jenkins → Manage Jenkins → System → Slack Notifications.
- 34. Configure the Slack workspace, channel and authentication token.
- 35. Store the Slack OAuth token in Jenkins as a Secret Text credential.

- a. ID: slack-token
  - b. Secret: <SLACK-OAUTH-TOKEN>
36. Create a new Jenkins Freestyle project.
- a. Job Name: Vprofile-1
37. Configure Source Code Management → Git and provide the repository URL.
- a. <https://github.com/betawins/Vprofile-1.git>
38. Configure the required Git branch.
- a. \*/master
39. Configure GitHub credentials if the repository requires authentication.
40. Configure Build Environment and enable the SonarQube environment option provided by the installed SonarQube Jenkins plugin.
41. Configure Build Steps → Execute SonarQube scanner analysis properties.
- i. sonar.projectKey=Sabear
  - ii. sonar.projectName=Sabear
  - iii. sonar.projectVersion=1.0
  - iv. sonar.sources=/var/lib/jenkins/workspace/\$JOB\_NAME/src/
  - v. sonar.binaries=target/classes/com/visualpathit/account/controller/
  - vi. sonar.junit.reportsPath=target/surefire-reports
  - vii. sonar.jacoco.reportPath=target/jacoco.exec
  - viii. sonar.java.binaries=src/main/java/com/visualpathit/account/
42. Configure Post-build Actions → Slack Notifications and select the #Jenkins-bot channel.
43. Configure Slack notifications for successful and failed builds.
44. Save the Jenkins Freestyle job configuration.
45. Click Build Now to execute the job.
46. Jenkins clones the Vprofile-1 repository into the workspace.
47. SonarQube Scanner analyzes the source code from the Jenkins workspace and sends the analysis results to the SonarQube server.
48. Verify the analysis result in:
- a. <http://<SONARQUBE-PUBLIC-IP>:9000>
49. Open Projects → Vprofile-1 and verify the Quality Gate result.
50. Verify that Jenkins sends the build status notification to:
- a. #Jenkins-bot

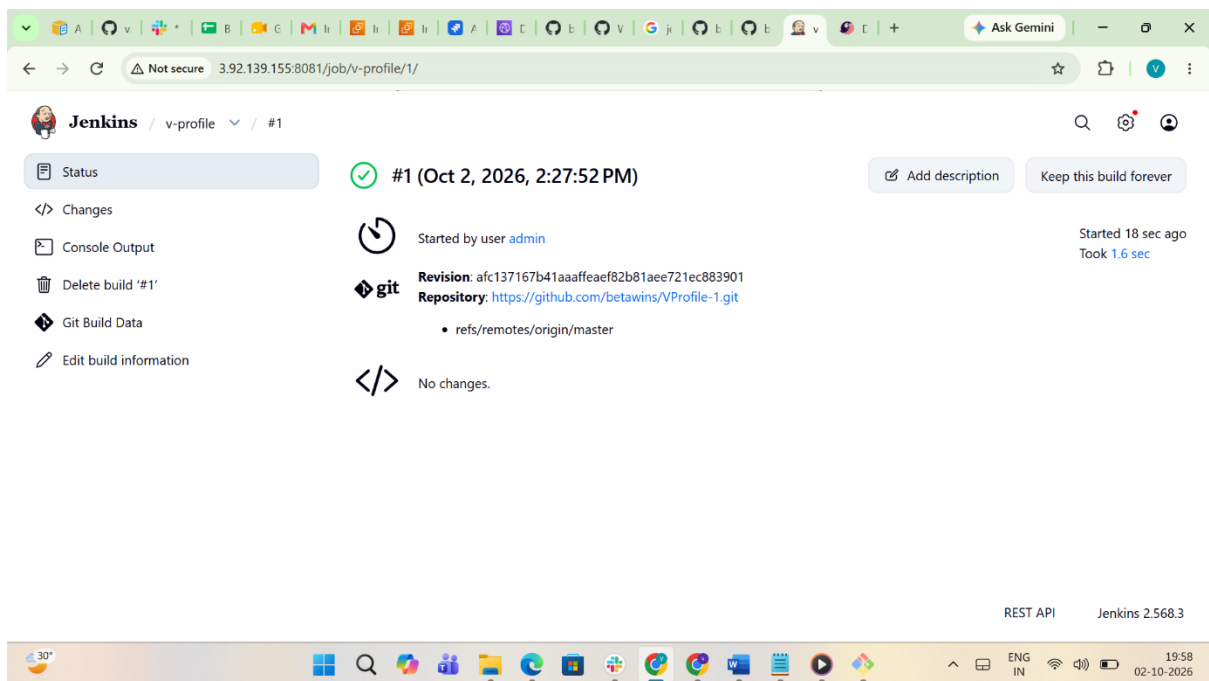
51. Introduce a temporary build failure and execute Build Now again.

a. exit 1

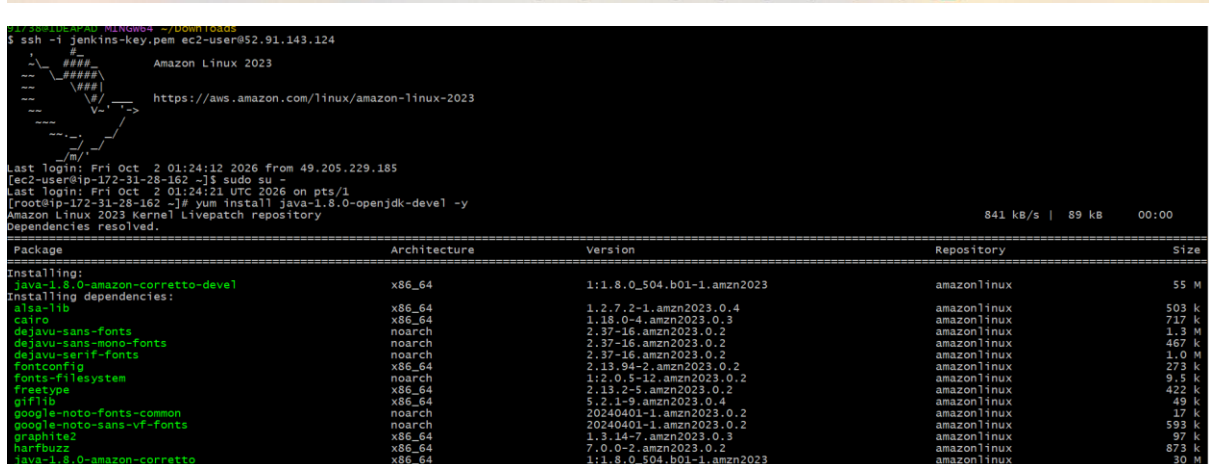
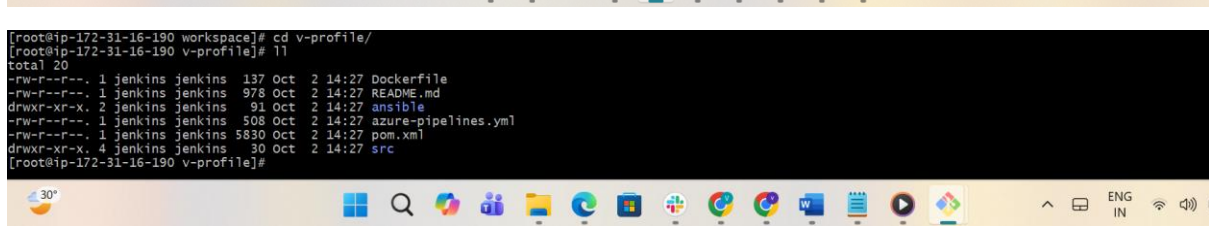
52. Verify that Jenkins reports the build as FAILED.

53. Verify that the failure notification is received in the Slack #Jenkins-bot channel.

54. Capture screenshots of the Jenkins job configuration, GitHub checkout, SonarQube analysis, SonarQube project/Quality Gate, Jenkins console output, successful Slack notification and failed Slack notification.



REST API Jenkins 2.568.3



```
[root@ip-172-31-16-190 workspace]# cd v-profile/
[root@ip-172-31-16-190 v-profile]# ll
total 20
-rw-r--r-- 1 jenkins jenkins 137 Oct 2 14:27 Dockerfile
-rw-r--r-- 1 jenkins jenkins 978 Oct 2 14:27 README.md
drwxr-xr-x 2 jenkins jenkins 91 Oct 2 14:27 ansible
-rw-r--r-- 1 jenkins jenkins 508 Oct 2 14:27 azure-pipelines.yml
-rw-r--r-- 1 jenkins jenkins 5830 Oct 2 14:27 pom.xml
drwxr-xr-x 4 jenkins jenkins 30 Oct 2 14:27 src
[root@ip-172-31-16-190 v-profile]#
```

```
$ ssh -i jenkins-key.pem ec2-user@52.91.143.124
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Fri Oct 2 01:24:12 2026 from 49.205.229.185
[ec2-user@ip-172-31-28-162 ~]$ sudo su -
[root@ip-172-31-28-162 ~]# yum install java-1.8.0-openjdk-devel -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
841 kB/s | 89 kB | 00:00

Package Architecture Version Repository Size
Installing:
java-1.8.0-amazon-corretto-devel x86_64 1:1.8.0_504.b01-1.amzn2023 amazonlinux 55 M
Installing dependencies:
alsa-lib x86_64 1.2.7.2-1.amzn2023.0.4 amazonlinux 503 k
cairo x86_64 1.18.0-4.amzn2023.0.3 amazonlinux 717 k
dejavu-sans-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 1.3 M
dejavu-sans-mono-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 467 k
dejavu-serif-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 1.0 M
fontconfig x86_64 2.13.94-2.amzn2023.0.2 amazonlinux 273 k
fontconfig-devel x86_64 1:2.13.94-2.amzn2023.0.2 amazonlinux 9.5 k
fontconfig-filesystem noarch 2.13.2-5.amzn2023.0.2 amazonlinux 422 k
giflib x86_64 5.2.1-9.amzn2023.0.4 amazonlinux 49 k
google-noto-fonts-common noarch 20240401-1.amzn2023.0.2 amazonlinux 17 k
google-noto-sans-vf-fonts noarch 20240401-1.amzn2023.0.2 amazonlinux 593 k
graphite2 x86_64 1.3.14-7.amzn2023.0.3 amazonlinux 97 k
harfbuzz x86_64 7.0.0-2.amzn2023.0.2 amazonlinux 873 k
java-1.8.0-amazon-corretto x86_64 1:1.8.0_504.b01-1.amzn2023 amazonlinux 30 M
```

```
MINGW64/c/Users/91738/Downloads
[ec2-user@ip-172-31-28-162 opt]$ sudo dnf install docker -y
Amazon Linux 2023 repository
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.

Package Architecture Version Repository Size
-----
Installing:
docker x86_64 25.0.16-1.amzn2023.0.4 amazonlinux 46 M
Installing dependencies:
container-selinux noarch 4.2.245-0-1.amzn2023 amazonlinux 58 k
containerd x86_64 2.2.7-1.amzn2023.0.1 amazonlinux 25 M
iptables-libs x86_64 1.8.8-3.amzn2023.0.2 amazonlinux 401 k
iptables-nft x86_64 1.8.8-3.amzn2023.0.2 amazonlinux 183 k
libcgrouper x86_64 3.0-1.amzn2023.0.1 amazonlinux 75 k
libnetfilter_conntrack x86_64 1.0.8-2.amzn2023.0.2 amazonlinux 58 k
libnetfilterlink x86_64 1.0.1-19.amzn2023.0.2 amazonlinux 30 k
libnftnl x86_64 1.2.2-2.amzn2023.0.2 amazonlinux 84 k
pigz x86_64 2.5-1.amzn2023.0.4 amazonlinux 83 k
runc x86_64 1.3.6-1.amzn2023.0.1 amazonlinux 4.0 M

Transaction Summary
-----
Install 11 Packages

Total download size: 77 M
Installed size: 287 M
Downloading Packages:
(1/11): container-selinux-2.245-0-1.amzn2023.noarch.rpm 2.2 MB/s | 58 kB 00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm 11 MB/s | 401 kB 00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm 4.0 MB/s | 183 kB 00:00
(4/11): libcgrouper-3.0-1.amzn2023.0.1.x86_64.rpm 3.5 MB/s | 75 kB 00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm 2.2 MB/s | 58 kB 00:00
(6/11): libnetfilterlink-1.0.1-19.amzn2023.0.2.x86_64.rpm 1.3 MB/s | 30 kB 00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm 3.0 MB/s | 84 kB 00:00
(8/11): pigz-2.5-1.amzn2023.0.4.x86_64.rpm 2.1 MB/s | 83 kB 00:00
(9/11): containerd-2.2.7-1.amzn2023.0.1.x86_64.rpm 75 MB/s | 25 MB 00:00
(10/11): runc-1.3.6-1.amzn2023.0.1.x86_64.rpm 33 MB/s | 4.0 MB 00:00
(11/11): docker-25.0.16-1.amzn2023.0.4.x86_64.rpm 61 MB/s | 46 MB 00:00
Total 95 MB/s | 77 MB 00:00

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction:
Preparing :
Installing : runc-1.3.6-1.amzn2023.0.1.x86_64
1/1
1/11
```

```
MINGW64/c/Users/91738/Downloads
pigz-2.5-1.amzn2023.0.4.x86_64 runc-1.3.6-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-28-162 opt]$ sudo systemctl start docker
[ec2-user@ip-172-31-28-162 opt]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-10-02 16:04:38 UTC; 12s ago
     TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
      Process: 58912 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
      Process: 58921 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
      Main PID: 58922 (dockerd)
        Tasks: 9
       Memory: 27.6M
          CPU: 234ms
      CGroup: /system.slice/docker.service
              └─58922 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Oct 02 16:04:37 ip-172-31-28-162.ec2.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Oct 02 16:04:37 ip-172-31-28-162.ec2.internal dockerd[58922]: time="2026-10-02T16:04:37.703853034Z" level=info msg="Starting up"
Oct 02 16:04:37 ip-172-31-28-162.ec2.internal dockerd[58922]: time="2026-10-02T16:04:37.740882196Z" level=info msg="Loading containers: start."
Oct 02 16:04:38 ip-172-31-28-162.ec2.internal dockerd[58922]: time="2026-10-02T16:04:38.057463914Z" level=info msg="Loading containers: done."
Oct 02 16:04:38 ip-172-31-28-162.ec2.internal dockerd[58922]: time="2026-10-02T16:04:38.077682336Z" level=info msg="Docker daemon" commit=6fdd0a6 containerd-snapshotter=fab
Oct 02 16:04:38 ip-172-31-28-162.ec2.internal dockerd[58922]: time="2026-10-02T16:04:38.077689696Z" level=info msg="Daemon has completed initialization"
Oct 02 16:04:38 ip-172-31-28-162.ec2.internal dockerd[58922]: time="2026-10-02T16:04:38.093106593Z" level=info msg="API listen on /run/docker.sock"
Oct 02 16:04:38 ip-172-31-28-162.ec2.internal systemd[1]: Started docker.service - Docker Application Container Engine.

[ec2-user@ip-172-31-28-162 opt]$ sudo docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-28-162 opt]$ sudo docker pull mysql:5.7
5.7: Pulling from library/mysql
20e4dcae4c68: Pull complete
1c56c3d4ce74: Pull complete
e9f03a1c24ce: Pull complete
88c3898c2015: Pull complete
6b93a940e7b5: Pull complete
90986bb8de6e: Pull complete
ae71319cb779: Pull complete
ffc89e9dfdd8: Pull complete
43a0e93819b: Pull complete
064b2d298fba: Pull complete
df9a4d85569b: Pull complete
Digest: sha256:4bc6bc9636d8443453676cae56536f4b8156d78bae03c0145cbe47c2aad73bb
Status: Downloaded newer image for mysql:5.7
docker.io/library/mysql:5.7
[ec2-user@ip-172-31-28-162 opt]$ sudo docker images | grep mysql
mysql 5.7 s107333e08a8 2 years ago 501MB
```

```
MINGW64/c/Users/91738/Downloads
[ec2-user@ip-172-31-28-162 opt]$ sudo docker volume create mysql57-data
mysql57-data
[ec2-user@ip-172-31-28-162 opt]$ sudo docker volume ls
DRIVER VOLUME NAME
local mysql57-data
[ec2-user@ip-172-31-28-162 opt]$ AC
[ec2-user@ip-172-31-28-162 opt]$ AC
[ec2-user@ip-172-31-28-162 opt]$ sudo ss -lntcp | grep ':3306'
[ec2-user@ip-172-31-28-162 opt]$ sudo docker run -d \
--name mysql57 \
--restart unless-stopped \
-e MYSQL_ROOT_PASSWORD='Root@123' \
-e MYSQL_DATABASE=sonar \
-e MYSQL_USER=sonar \
-e MYSQL_PASSWORD='sonar@123' \
-v mysql57-data:/var/lib/mysql \
-p 3306:3306 \
mysql:5.7
7b6946518ba44f95ec3c7dc9d3b2907f5896a43c9f5ac9c8f8c3a1915d78526e
[ec2-user@ip-172-31-28-162 opt]$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
7b6946518ba4 mysql:5.7 "docker-entrypoint.s..." 34 seconds ago Up 33 seconds 0.0.0.0:3306->3306/tcp, :::3306->3306/tcp, 33060/tcp mysql57
```

```
MINGW64/C:/Users/91738/Downloads
[ec2-user@ip-172-31-28-162 opt]$ sudo docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
7b6946518ba4   mysql:5.7      "docker-entrypoint.s..." 3 minutes ago  Up 3 minutes  0.0.0.0:3306->3306/tcp, :::3306->3306/tcp, 33060/tcp  mysql157
[ec2-user@ip-172-31-28-162 opt]$ sudo docker exec -it mysql157 mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 180
Server version: 5.7.44 MySQL Community Server (GPL)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SELECT VERSION();
+-----+
| VERSION() |
+-----+
| 5.7.44    |
+-----+
1 row in set (0.00 sec)

mysql> show databases;
ERROR 1064 (42000): you have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'databases' at line 1
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql      |
| performance_schema |
| sonar      |
| sys        |
+-----+
5 rows in set (0.00 sec)

mysql> exit;
Bye
[ec2-user@ip-172-31-28-162 opt]$ sudo docker exec mysql157 mysql -u sonar -p sonar -e "SELECT DATABASE();"
Enter password: ERROR 1045 (28000): Access denied for user 'sonar'@'localhost' (using password: NO)
[ec2-user@ip-172-31-28-162 opt]$ sudo docker exec -it mysql157 mysql -u sonar -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 182
```

```
1 row in set (0.00 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| sonar      |
+-----+
2 rows in set (0.00 sec)

mysql> SHOW TABLES;
ERROR 1046 (3D000): No database selected
mysql> USE sonar;
Database changed
mysql> SHOW TABLES;
Empty set (0.00 sec)

mysql> SHOW GRANTS;
+-----+
| Grants for sonar@% |
+-----+
| GRANT USAGE ON *.* TO 'sonar'@'%' |
| GRANT ALL PRIVILEGES ON `sonar`.* TO 'sonar'@'%' |
+-----+
2 rows in set (0.00 sec)
```

```
MINGW64/c/Users/91738/Downloads
-bash: $'\r': command not found
[root@ip-172-31-28-162 opt]# wget https://repo1.maven.org/maven2/org/sonarsource/sonarqube/sonar-application/6.6/sonar-application-6.6.zip -O sonarqube-6.6.zip
--2026-10-02 15:00:31-- https://repo1.maven.org/maven2/org/sonarsource/sonarqube/sonar-application/6.6/sonar-application-6.6.zip
Resolving repo1.maven.org (repo1.maven.org)... 104.18.19.12, 104.18.18.12, 2606:4700:6812:130c, ...
Connecting to repo1.maven.org (repo1.maven.org)|104.18.19.12|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 164350442 (157M) [application/zip]
Saving to: 'sonarqube-6.6.zip'

sonarqube-6.6.zip                               100%[=====] 156.74M  73.8MB/s   in 2.1s

2026-10-02 15:00:34 (73.8 MB/s) - 'sonarqube-6.6.zip' saved [164350442/164350442]

[root@ip-172-31-28-162 opt]# ll
total 160500
drwxr-xr-x. 4 root root      33 Sep 28 23:38 aws
-rw-r--r--. 1 root root 164350442 Oct 20 2017 sonarqube-6.6.zip
[root@ip-172-31-28-162 opt]# unzip sonarqube-6.6.zip
Archive:  sonarqube-6.6.zip
  creating: sonarqube-6.6/
  creating: sonarqube-6.6/lib/
  inflating: sonarqube-6.6/lib/sonar-application-6.6.jar
  creating: sonarqube-6.6/lib/jsw/
  inflating: sonarqube-6.6/lib/jsw/wrapper-3.2.3.jar
  creating: sonarqube-6.6/lib/server/
  inflating: sonarqube-6.6/lib/server/sonar-server-6.6.jar
  inflating: sonarqube-6.6/lib/server/jcl-over-slf4j-1.7.24.jar
  inflating: sonarqube-6.6/lib/server/logback-access-1.1.7.jar
  inflating: sonarqube-6.6/lib/server/tomcat-embed-core-8.5.16.jar
  inflating: sonarqube-6.6/lib/server/gson-2.3.1.jar
  inflating: sonarqube-6.6/lib/server/sonar-core-6.6.jar
  inflating: sonarqube-6.6/lib/server/picocontainer-2.15.jar
  inflating: sonarqube-6.6/lib/server/protobuf-java-3.0.0-beta-2.jar
  inflating: sonarqube-6.6/lib/server/sonar-classloader-1.0.jar
  inflating: sonarqube-6.6/lib/server/sonar-plugin-api-6.6.jar
  inflating: sonarqube-6.6/lib/server/woodstox-core-5.4.0.jar
  inflating: sonarqube-6.6/lib/server/stax2-api-3.1.4.jar
  inflating: sonarqube-6.6/lib/server/staxmate-2.0.1.jar
  inflating: sonarqube-6.6/lib/server/sonar-update-center-common-1.18.0.487.jar
  inflating: sonarqube-6.6/lib/server/log4j-over-slf4j-1.7.24.jar
  inflating: sonarqube-6.6/lib/server/sonar-db-dao-6.6.jar
  inflating: sonarqube-6.6/lib/server/sonar-db-core-6.6.jar
  inflating: sonarqube-6.6/lib/server/commons-csv-1.4.jar
  inflating: sonarqube-6.6/lib/server/sonar-scanner-protocol-6.6.jar
  inflating: sonarqube-6.6/lib/server/sonar-lz4-1.3.0.jar
  inflating: sonarqube-6.6/lib/server/mybatis-3.4.4.jar
  inflating: sonarqube-6.6/lib/server/commons-dbcp-1.4.jar
  inflating: sonarqube-6.6/lib/server/commons-pool-1.5.4.jar
```

```
  inflating: sonarqube-6.6/elasticsearch/VERSION.txt
  inflating: sonarqube-6.6/elasticsearch/config/elasticsearch.yml
  inflating: sonarqube-6.6/elasticsearch/config/jvm.options
  inflating: sonarqube-6.6/elasticsearch/config/log4j2.properties
  inflating: sonarqube-6.6/bin/jsw-license/LICENSE.txt
  inflating: sonarqube-6.6/extensions/plugins/README.txt
  inflating: sonarqube-6.6/extensions/jdbc-driver/oracle/README.txt
  inflating: sonarqube-6.6/COPYING
  inflating: sonarqube-6.6/temp/README.txt
  inflating: sonarqube-6.6/data/README.txt
  creating: sonarqube-6.6/logs/
  creating: sonarqube-6.6/elasticsearch/plugins/
[root@ip-172-31-28-162 opt]# ls
aws sonarqube-6.6 sonarqube-6.6.zip
[root@ip-172-31-28-162 opt]# mv /opt/sonarqube-6.6 /opt/sonar
[root@ip-172-31-28-162 opt]# ls
aws sonar sonarqube-6.6.zip
```

```
[root@ip-172-31-28-162 opt]# cd sonar/
[root@ip-172-31-28-162 sonar]# ls
COPYING bin conf data elasticsearch extensions lib logs temp web
[root@ip-172-31-28-162 sonar]# cd conf/
[root@ip-172-31-28-162 conf]# ls
sonar.properties wrapper.conf
[root@ip-172-31-28-162 conf]# vi sonar.properties
[root@ip-172-31-28-162 conf]# exit
logout
[ec2-user@ip-172-31-28-162 ~]$ cd /opt/sonar/
[ec2-user@ip-172-31-28-162 sonar]$ ls
COPYING bin conf data elasticsearch extensions lib logs temp web
[ec2-user@ip-172-31-28-162 sonar]$ cd bin/
[ec2-user@ip-172-31-28-162 bin]$ ls
jsw-license linux-x86-32 linux-x86-64 macosx-universal-64 windows-x86-32 windows-x86-64
[ec2-user@ip-172-31-28-162 bin]$ cd /opt/
[ec2-user@ip-172-31-28-162 opt]$ ls
aws sonar sonarqube-6.6.zip
[ec2-user@ip-172-31-28-162 opt]$ ll
total 160500
drwxr-xr-x. 4 root root      33 Sep 28 23:38 aws
drwxr-xr-x. 11 root root      141 Oct 20 2017 sonar
-rw-r--r--. 1 root root 164350442 Oct 20 2017 sonarqube-6.6.zip
[ec2-user@ip-172-31-28-162 opt]$ sudo chown -R
chown: missing operand
Try 'chown --help' for more information.
[ec2-user@ip-172-31-28-162 opt]$ sudo chown -R ec2-user:ec2-user sonar
[ec2-user@ip-172-31-28-162 opt]$ ll
total 160500
drwxr-xr-x. 4 root      root      33 Sep 28 23:38 aws
drwxr-xr-x. 11 ec2-user ec2-user  141 Oct 20 2017 sonar
-rw-r--r--. 1 root      root      164350442 Oct 20 2017 sonarqube-6.6.zip
[ec2-user@ip-172-31-28-162 opt]$ cd sonar/bin/
[ec2-user@ip-172-31-28-162 bin]$ ls
jsw-license linux-x86-32 linux-x86-64 macosx-universal-64 windows-x86-32 windows-x86-64
[ec2-user@ip-172-31-28-162 bin]$ cd linux-x86-64/
[ec2-user@ip-172-31-28-162 linux-x86-64]$ ls
lib sonar.sh wrapper
[ec2-user@ip-172-31-28-162 linux-x86-64]$ ./sonar.sh start
Starting SonarQube...
Started SonarQube.
[ec2-user@ip-172-31-28-162 linux-x86-64]$ ./sonar.sh status
SonarQube is running (54773).
```

```
# User credentials.
# Permissions to create tables, indices and triggers must be granted to JDBC user.
# The schema must be created first.
sonar.jdbc.username=sonar
sonar.jdbc.password=sonar@123

#----- Embedded Database (default)
# H2 embedded database server listening port, defaults to 9092
#sonar.embeddedDatabase.port=9092

#----- MySQL 5.6 or greater
# Only InnoDB storage engine is supported (not myISAM).
# Only the bundled driver is supported. It can not be changed.
sonar.jdbc.url=jdbc:mysql://localhost:3306/sonar?useUnicode=true&characterEncoding=utf8&rewriteBatchedStatements=true&useConfigs=maxPerformance&useSSL=false

#----- Oracle 11g/12c
# The Oracle JDBC driver must be copied into the directory extensions/jdbc-driver/oracle/.
# Only the thin client is supported, and only the versions 11.2.x or 12.2.x must be used. See
# https://docs.sonarqube.org/latest/setup/ for more details.
```

```
# Binding IP address. For servers with more than one IP address, this property specifies which
# address will be used for listening on the specified ports.
# By default, ports will be used on all IP addresses associated with the server.
sonar.web.host=0.0.0.0

# Web context. When set, it must start with forward slash (for example /sonarqube).
# The default value is root context (empty value).
sonar.web.context=/sonar
# TCP port for incoming HTTP connections. Default value is 9000.
#sonar.web.port=9000

# The maximum number of connections that the server will accept and process at any given time.
# When this number has been reached, the server will not accept any more connections until
# the number of connections falls below this value. The operating system may still accept connections
# based on the sonar.web.connections.acceptCount property. The default value is 50.
#sonar.web.http.maxThreads=50
```

The screenshot shows the SonarQube web interface in a browser. The address bar indicates the URL is 52.91.143.124:9000/sonar/projects. The interface has a top navigation bar with tabs for 'Projects', 'Issues', 'Rules', 'Quality Profiles', 'Quality Gates', and 'Administration'. A search bar is present on the right. On the left, there are filters for 'Quality Gate' (Passed, Warning, Failed) and 'Reliability' (A, B, C, D, E). The main content area shows 'Perspective: Overall Status' and 'Sort by: Name'. Below this, it states 'Once you analyze some projects, they will show up here.' and '0 of 0 shown'. The footer indicates 'SonarQube™ technology is powered by SonarSource SA' and 'Version 6.6 (build 32724) - LGPL v3 - Community - Documentation - Get Support - Plugins - Web API - About'.

sonarqube Projects Issues Rules Quality Profiles Quality Gates Search for projects, sub-projects and files... Log in

## Continuous Code Quality

[Log in](#) [Read documentation](#)

0 Projects Analyzed

- 0 Bugs
- 0 Vulnerabilities
- 0 Code Smells

### Multi-Language

20+ programming languages are supported by SonarQube thanks to our in-house code analyzers, including:

|        |        |      |        |        |      |       |              |             |     |
|--------|--------|------|--------|--------|------|-------|--------------|-------------|-----|
| Java   | C/C++  | C#   | COBOL  | ABAP   | HTML | RPG   | JavaScript   | Objective C | XML |
| VB.NET | PL/SQL | Flex | Python | Groovy | PHP  | Swift | Visual Basic | PL/I        |     |

### Quality Model

\* Bug track ends that is demonstrated by a... \* Vulnerabilities are... \* Code Smells will...  
27° 21:58 02-10-2026

Jenkins / Manage Jenkins / Plugins

## Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

## Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

Joda Time API ⓘ Downloaded Successfully. Will be activated during the next boot

SonarQube Scanner ✓ Success

Loading plugin extensions ✓ Success

→ [Go back to the top page](#)  
(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

REST API Jenkins 2.568.3

28° 22:04 02-10-2026



3.92.139.155:8081/manage/configureTools/

### Jenkins

Manage Jenkins / Tools

SonarQube Scanner

Name  
sonar\_scanner

SONAR\_RUNNER\_HOME  
/opt/sonar\_scanner

☐ Install automatically ?

+ Add SonarQube Scanner

Ant installations

Save Apply

25° ENG IN 22:33 02-10-2026

3.92.139.155:8081/job/v-profile/configure

### Jenkins

v-profile / Configure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Path to project properties ?

Analysis properties ?  
sonar.projectKey=Sabear  
sonar.projectName=Sabear  
sonar.projectVersion=1.0  
sonar.sources=/var/lib/jenkins/workspace/\$JOB\_NAME/src/  
sonar.binaries=target/classes/com/visualpathit/account/controller/  
sonar.junit.reportsPath=target/surefire-reports

Additional arguments ?

JVM Options ?

Save Apply

25° ENG IN 22:39 02-10-2026

Jenkins / v-profile

Status

Changes

Workspace

Build Now

Delete Project

Configure

Rename

SonarQube

v-profile

v-profile-app

SonarQube

Permalinks

- Last build (#1), 2 hr 42 min ago
- Last stable build (#1), 2 hr 42 min ago
- Last successful build (#1), 2 hr 42 min ago
- Last completed build (#1), 2 hr 42 min ago

Builds >

Filter

Today

#1 2:27 PM

25°

22:40 02-10-2026

Jenkins / v-profile / #2

INFO: CPD calculation finished

INFO: Analysis report generated in 175ms, dir size=109 KB

INFO: Analysis reports compressed in 56ms, zip size=48 KB

INFO: Analysis report uploaded in 184ms

INFO: ANALYSIS SUCCESSFUL, you can browse <http://52.91.143.124:9000/sonar/dashboard/index/Sabear>

INFO: Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report

INFO: More about the report processing at <http://52.91.143.124:9000/sonar/api/ce/task?id=AaD9mGyf48n6JUCFFXO>

INFO: Task total time: 6.436 s

INFO: -----

INFO: EXECUTION SUCCESS

INFO: -----

INFO: Total time: 9.086s

INFO: Final Memory: 8M/30M

INFO: -----

Finished: SUCCESS

Jenkins 2.568.3

25°

22:41 02-10-2026

sonarqube / Projects

Quality Gate

Reliability ( Bugs )

Security ( Vulnerabilities )

Perspective: Overall Status

Sort by: Name

Search by project name or key

1 projects

Sabear

Passed

0 Bugs

0 Vulnerabilities

0 Code Smells

0.0% Duplications

148 XML

Last analysis: October 2, 2026 at 10:40 PM

1 of 1 shown

SonarQube™ technology is powered by SonarSource SA

Version 6.6 (build 32724) - LGPL v3 - Community - Documentation - Get Support - Plugins - Web API - About

25°

22:41 02-10-2026

```
ec2-user@ip-172-31-28-162:/opt/sonar/bin/linux-x86-64
mysql> use sonar;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
Tables_in_sonar
+-----+
active_rule_parameters
active_rules
ce_activity
ce_queue
ce_scanner_context
ce_task_characteristics
ce_task_input
default_profiles
duplications_index
es_queue
events
file_sources
group_roles
groups
groups_users
internal_properties
issue_changes
issues
loaded_templates
manual_measures
metrics
notifications
org_profiles
organization_members
organizations
perm_templates_groups
perm_templates_users
perm_tpl_characteristics
permission_templates
plugins
project_branches
project_links
project_measures
project_profiles
projects
properties
qprofile_changes
```

Jenkins / Manage Jenkins / Plugins

### Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

### Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

Apache HttpComponents Client 5.x API Success

Slack Notification Success

Loading plugin extensions Success

→ [Go back to the top page](#)  
(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

slack api

### Your Apps

Use our APIs to build your app

Documentation Tutorials Your Apps

#### Create AI agent

**Help desk**  
Includes tools for knowledge search, tickets, and status checks.

**Starter agent**  
Starts with minimal tools and add your own tools and logic from there.

Agent name

Letters, spaces, numbers, dashes, and parentheses only. You can change this later.

Workspace

jenkins-bot

This can't be changed later. If you leave this workspace, you'll lose access to this app.

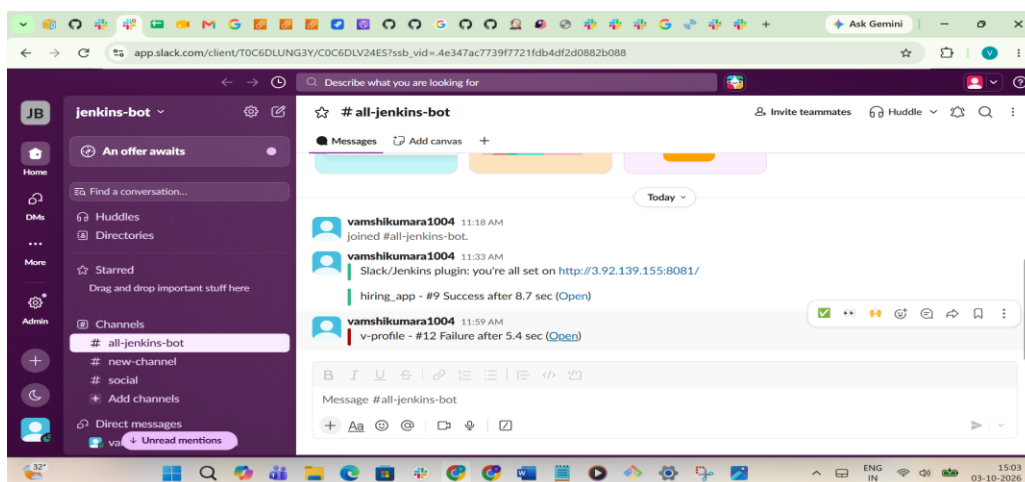
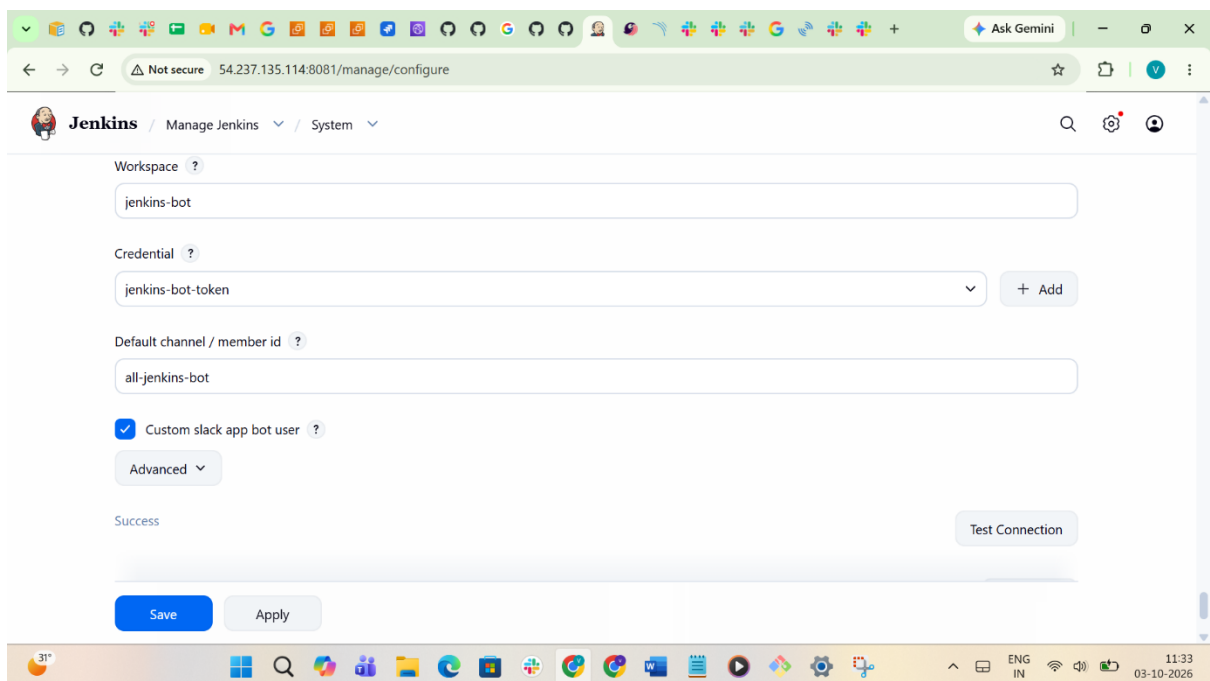
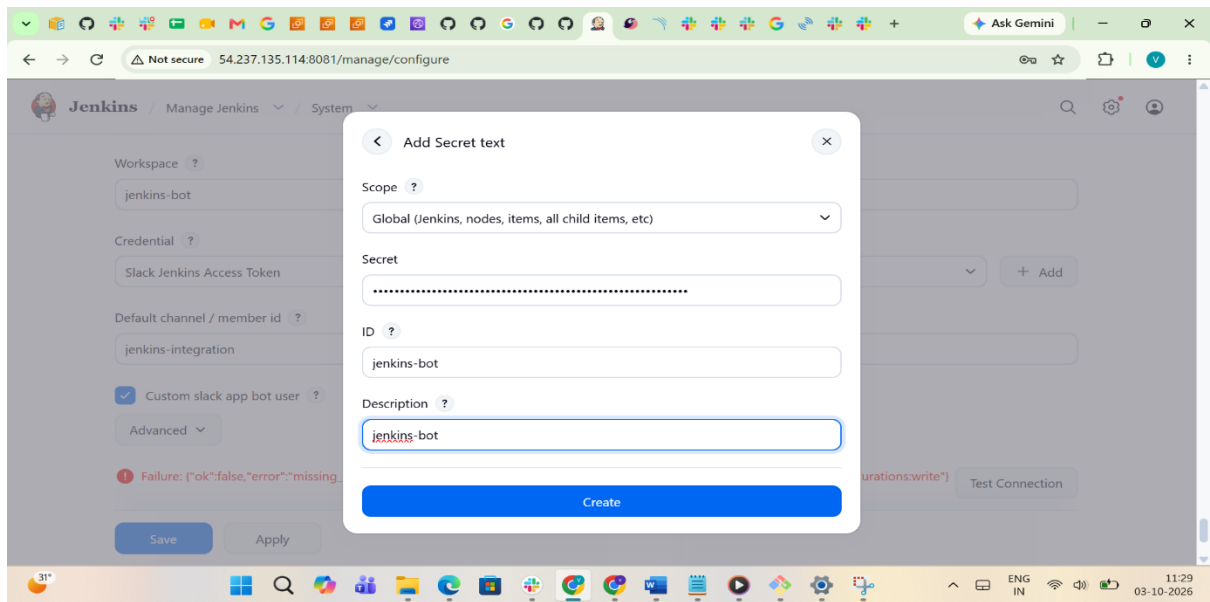
Step 1 of 2

BackReview Agent

Workspace

TechieHorizon

Copy



## 2. Create one Jenkins job using the below code and create three stages.

- **stage1: Git clone to download the source code.**
- **stage2: Sonarqube Integration to check the quality of code.**
- **stage3: Slack Integration to send the alerts to slack.**

**Source Code:** <https://github.com/betawins/hiring-app.git>

### Implementation – Jenkins Freestyle Job with GitHub, SonarQube and Slack

1. Login to the Jenkins EC2 server and switch to the root user. `sudo su -i`
2. Verify Jenkins service and configured port. `systemctl status jenkins`
  - a. `ss -lntp | grep 8081`
3. Access Jenkins using the configured URL.
  - a. `http://<JENKINS-PUBLIC-IP>:8081`
4. Login to Jenkins and install the required plugins from Manage Jenkins → Plugins:
  - a. Git
  - b. SonarQube Scanner
  - c. Slack Notification
5. Login to the SonarQube EC2 server and verify the installed Java version.
  - a. `java -version`
6. Install Java 8 required for the selected SonarQube version.
  - a. `dnf install -y java-1.8.0-amazon-corretto`
7. Verify Java 8 installation.
  - a. `java -version`
8. Install Docker on the SonarQube server.
  - a. `dnf install -y docker`
  - b. `systemctl start docker`
  - c. `systemctl enable docker`
9. Pull the MySQL 5.7 Docker image.
  - a. `docker pull mysql:5.7`
10. Create and start the MySQL 5.7 container with the SonarQube database and user.
  - a. `docker run -d --name mysql57 \`

- b. -e MYSQL\_ROOT\_PASSWORD=Root@123 \
- c. -e MYSQL\_DATABASE=sonar \
- d. -e MYSQL\_USER=sonar \
- e. -e MYSQL\_PASSWORD=Sonar@123 \
- f. -p 3306:3306 mysql:5.7

11. Verify that the MySQL container is running.

- a. docker ps

12. Login to MySQL and verify the sonar database and sonar user.

- a. docker exec -it mysql57 mysql -u root -p
- b. SHOW DATABASES;
- c. SELECT user,host FROM mysql.user;

13. Pull the required SonarQube Docker image.

- a. docker pull sonarqube:<version>

14. Create and start the SonarQube Docker container and connect it to the MySQL container.

```
docker run -d \
--name sonarqube \
--network sonar-network \
-p 9000:9000 \
-e SONAR_JDBC_URL=jdbc:mysql://mysql57:3306/sonar \
-e SONAR_JDBC_USERNAME=sonar \
-e SONAR_JDBC_PASSWORD=Sonar@123 \
sonarqube:<version>
```

- a. docker run -itd -p 9002:9000 sonarqube

15. Verify that the MySQL and SonarQube containers are running.

- a. docker ps

16. Verify the SonarQube container logs.

- a. docker logs sonarqube

17. Monitor the SonarQube container logs until the server starts successfully.

- a. docker logs -f sonarqube

18. Verify that SonarQube is running on port 9002.

- a. `ss -lntp | grep 9002`
- 19. Access SonarQube using:
  - a. `http://<SONARQUBE-PUBLIC-IP>:9002`
- 20. Login to SonarQube and create an authentication token from My Account → Security → Tokens.
- 21. Login to Jenkins and configure the SonarQube server under Manage Jenkins → System.
- 22. Add the SonarQube server with the following details:
  - a. Name: My-SonarQube
  - b. Server URL: `http://<SONARQUBE-PUBLIC-IP>:9002`
  - c. Credential: sonarqube-token
- 23. Create the SonarQube token in Jenkins as a Secret Text credential.
  - a. ID: sonarqube-token
  - b. Secret: <SONARQUBE-TOKEN>
- 24. Configure SonarQube Scanner under Manage Jenkins → Tools → SonarQube Scanner installations.
- 25. Configure the SonarQube Scanner properties with the SonarQube server URL.
  - a. `sonar.host.url=http://<SONARQUBE-PUBLIC-IP>:9002`
- 26. Create a Slack channel named:
  - a. #Jenkins-bot
- 27. Configure Slack integration under Jenkins → Manage Jenkins → System → Slack Notifications.
- 28. Configure the Slack workspace, channel and authentication token.
- 29. Store the Slack OAuth token in Jenkins as a Secret Text credential.
  - a. ID: slack-token
  - b. Secret: <SLACK-OAUTH-TOKEN>
- 30. Create a new Jenkins Freestyle project.
  - a. Job Name: hiring-app
- 31. Configure Source Code Management → Git and provide the repository URL.
  - a. `https://github.com/betawins/hiring-app.git`
- 32. Configure the required Git branch.
  - a. \*/main

33. Configure GitHub credentials if the repository requires authentication.
34. Configure Build Environment and enable the SonarQube environment option provided by the installed SonarQube Jenkins plugin.
35. Configure Build Steps → Execute SonarQube scanner analysis properties.

```
sonar.projectKey=Test
sonar.sources=.
```
36. Configure Post-build Actions → Slack Notifications and select the #Jenkins-bot channel.
37. Configure Slack notifications for successful and failed builds.
38. Save the Jenkins Freestyle job configuration.
39. Click Build Now to execute the job.
40. Jenkins clones the hiring-app repository into the workspace.
41. SonarQube Scanner analyzes the source code from the Jenkins workspace and sends the analysis results to the SonarQube server.
42. Verify the analysis result in:
  - a. <http://<SONARQUBE-PUBLIC-IP>:9002>
43. Open Projects → hiring-app and verify the Quality Gate result.
44. Verify that Jenkins sends the build status notification to:
  - a. #Jenkins-bot
45. Introduce a temporary build failure and execute Build Now again.
  - a. `exit 1`
46. Verify that Jenkins reports the build as FAILED.
47. Verify that the failure notification is received in the Slack #Jenkins-bot channel.
- 48.** Capture screenshots of the Jenkins job configuration, GitHub checkout, SonarQube analysis, SonarQube project/Quality Gate, Jenkins console output, successful Slack notification and failed Slack notification.

Jenkins / hiring\_app / Configure

## Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Git ?

Repositories ?

Repository URL ?

<https://github.com/betawins/hiring-app.git>

Credentials ?

- none -

+ Add

Advanced

+ Add Repository

Branches to build ?

Save Apply

Jenkins / hiring\_app / #1

Status

Changes

Console Output

Delete build '#1'

Git Build Data

Edit build information

#1 (Oct 3, 2026, 4:02:48 AM)

Add description

Keep this build forever

Started by user admin

Started 6.7 sec ago  
Took 1 sec

Revision: 9405b107fdffdf68e69fe953ea1f9669784f9d6a  
Repository: <https://github.com/betawins/hiring-app.git>

- refs/remotes/origin/main

No changes.

REST API Jenkins 2.568.3

Jenkins / hiring\_app / #1

Git Build Data

Edit build information

```
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/betawins/hiring-app.git
> git init /var/lib/jenkins/workspace/hiring_app # timeout=10
Fetching upstream changes from https://github.com/betawins/hiring-app.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/betawins/hiring-app.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/betawins/hiring-app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 9405b107fdffdf68e69fe953ea1f9669784f9d6a (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 9405b107fdffdf68e69fe953ea1f9669784f9d6a # timeout=10
Commit message: "Update README.md"
First time build. Skipping changelog.
Finished: SUCCESS
```

```
[ec2-user@ip-172-31-28-162 linux-x86-64]$ systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
   Active: active (running) since Sat 2026-10-03 03:51:27 UTC; 19min ago
   TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
   Process: 3052 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
   Process: 3053 ExecStartPre=/usr/libexec/docker/docker-setup-run-times.sh (code=exited, status=0/SUCCESS)
   Main PID: 3054 (dockerd)
     Tasks: 26
    Memory: 106.6M
       CPU: 1.027s
    CGroup: /system.slice/docker.service
            └─3054 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536
              └─3246 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 3306 -container-ip 172.17.0.2 -container-port 3306
                └─3252 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 3306 -container-ip 172.17.0.2 -container-port 3306

Oct 03 03:51:25 ip-172-31-28-162.ec2.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Oct 03 03:51:25 ip-172-31-28-162.ec2.internal dockerd[3054]: time="2026-10-03T03:51:25.873279530Z" level=info msg="Starting up"
Oct 03 03:51:25 ip-172-31-28-162.ec2.internal dockerd[3054]: time="2026-10-03T03:51:25.983713542Z" level=info msg="[graphdriver] using prior storage driver: overlay2"
Oct 03 03:51:26 ip-172-31-28-162.ec2.internal dockerd[3054]: time="2026-10-03T03:51:26.034253238Z" level=info msg="Loading containers: start."
Oct 03 03:51:26 ip-172-31-28-162.ec2.internal dockerd[3054]: time="2026-10-03T03:51:26.638325214Z" level=info msg="Default bridge (docker0) is assigned with an IP address 172.17.0.2"
Oct 03 03:51:27 ip-172-31-28-162.ec2.internal dockerd[3054]: time="2026-10-03T03:51:27.048472556Z" level=info msg="Loading containers: done."
Oct 03 03:51:27 ip-172-31-28-162.ec2.internal dockerd[3054]: time="2026-10-03T03:51:27.088148936Z" level=info msg="Docker daemon" commit=6fddf0a6 containerd-snapshotter=fal
Oct 03 03:51:27 ip-172-31-28-162.ec2.internal dockerd[3054]: time="2026-10-03T03:51:27.088929244Z" level=info msg="Daemon has completed initialization"
Oct 03 03:51:27 ip-172-31-28-162.ec2.internal dockerd[3054]: time="2026-10-03T03:51:27.119595713Z" level=info msg="API listen on /run/docker.sock"
Oct 03 03:51:27 ip-172-31-28-162.ec2.internal systemd[1]: Started docker.service - Docker Application Container Engine.

[ec2-user@ip-172-31-28-162 linux-x86-64]$ sudo su -
Last login: Fri Oct 2 14:45:40 UTC 2026 on pts/1
[root@ip-172-31-28-162 ~]# docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
a60ef8ec184e: Pull complete
96fe3bd0148e: Pull complete
54929e687af: Pull complete
4f4fb700ef54: Pull complete
0a66c41acdb: Pull complete
065adb12f4f5: Pull complete
03217b7e6be2: Pull complete
Digest: sha256:c0f1160ccfa4435db4168c2d7df69af3c3ea7bfeb87395613014048fb33a68c1
Status: Downloaded newer image for sonarqube:latest
docker.io/library/sonarqube:latest
[root@ip-172-31-28-162 ~]#
```

```
[root@ip-172-31-28-162 ~]# docker run -itd -p 9002:9000 sonarqube
97949015912ee1fab028e6cd288af0c88be8dca2136747c13f8ddaef8c696bb0
[root@ip-172-31-28-162 ~]# docker ps
```

| CONTAINER ID | IMAGE     | COMMAND                  | CREATED        | STATUS        | PORTS                                     | NAMES         |
|--------------|-----------|--------------------------|----------------|---------------|-------------------------------------------|---------------|
| 97949015912e | sonarqube | "/opt/sonarqube/dock..." | 39 seconds ago | Up 38 seconds | 0.0.0.0:9002->9000/tcp, :::9002->9000/tcp | sweet_hawking |
| 7b6946518ba4 | mysql:5.7 | "docker-entrypoint.s..." | 12 hours ago   | Up 23 minutes | 0.0.0.0:3306->3306/tcp, :::3306->3306/tcp | mysql157      |

Ask Gemini

34.235.166.178:9002/projects/create

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine.

SonarQube community

Projects Issues Rules Quality profiles Quality gates Administration More

## How do you want to create your project?

Do you want to benefit from all of SonarQube Community Build's features (like repository import and Pull Request decoration)?

### Create your project from your favorite DevOps platform.

First, you need to set up a DevOps platform configuration.

Import from Azure DevOps

Setup

Import from Bitbucket Cloud

Setup

Import from Bitbucket Data Center

Set...

Import from GitHub

Setup

Import from GitLab

Setup

Are you just testing or have an advanced use-case?

27°

Windows Taskbar

ENG IN

09:49 03-10-2026



The screenshot shows the Jenkins 'Configure' page for a job named 'hiring\_app'. The 'Build Steps' tab is selected, displaying a shell command block. The command sets environment variables for Java and SonarQube, runs 'java -version', and then executes 'sonar-scanner' with specific flags for host URL, token, project key, sources, and workspace.

```
export JAVA_HOME=/usr/lib/jvm/java-21-amazon-corretto.x86_64
export PATH=$JAVA_HOME/bin:$PATH

echo "=====
echo "Java used by SonarScanner"
echo "=====

java -version

echo "JAVA_HOME=$JAVA_HOME"

echo "=====
echo "Running SonarQube Scanner"
echo "=====

/opt/sonar_scanner/bin/sonar-scanner \
-Dsonar.host.url=http://34.235.166.178:9002/ \
-Dsonar.token='sqa_fd5edbie2b0421b508ba434a03151bd4e80882ee' \
-Dsonar.projectKey=Test \
-Dsonar.sources=. \
-Dsonar.projectBaseDir='${WORKSPACE}'
```

At the bottom of the command block, there are 'Save' and 'Apply' buttons.

Jenkins / hiring\_app / #7

```
INFO: SCM revision ID '9405b107dfdf68e69fe953eaf9669784f9d6a'
INFO: Analysis report generated in 418ms, dir size=294.7 kB
INFO: Analysis report compressed in 30ms, zip size=35.3 kB
INFO: Analysis report uploaded in 250ms
INFO: ANALYSIS SUCCESSFUL, you can find the results at: http://34.235.166.178:9002/dashboard?id=Test
INFO: Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
INFO: More about the report processing at http://34.235.166.178:9002/api/ce/task?id=0151877c-683c-4c84-8526-a6e1cc945f62
INFO: Analysis total time: 6.728 s
INFO: -----
INFO: EXECUTION SUCCESS
INFO: -----
INFO: Total time: 9.617s
INFO: Final Memory: 17M/60M
INFO: -----
Finished: SUCCESS
```

Jenkins 2.568.3

34.235.166.178:9002/projects

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine.

The way in which security, reliability, and maintainability counts and ratings are calculated has changed. [Learn more in SonarQube documentation](#)

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My Favorites All

Search projects Perspective Overall Status Sort by Name 1 project(s)

Test Public

Last analysis: 3 minutes ago - 45 Lines of Code - XML, Docker, ...

Passed

2 3 0 — — 0.0%

Security Reliability Maintainability Hotspots Reviewed Coverage Duplications

1 of 1 shown

SonarQube™ technology is powered by SonarSource Sàrl

Community Build v26.9.0.129388

LGPL v3 Community Documentation Plugins Web API

root@ip-172-31-28-162:~

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> use sonar;
Reading table information for completion of table and column names
You can turn off this feature with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_sonar |
+-----+
| active_rule_parameters |
| active_rules |
| ce_activity |
| ce_queue |
| ce_scanner_context |
| ce_task_characteristics |
| ce_task_input |
| default_profiles |
| duplications_index |
| es_queue |
| events |
| file_sources |
| group_roles |
| groups |
| groups_users |
| internal_properties |
| issue_changes |
| issues |
| loaded_templates |
| manual_measures |
| metrics |
| notifications |
| org_profiles |
| organization_members |
| organizations |
| perm_templates_groups |
| perm_templates_users |
| perm_tpl_characteristics |
| permission_templates |
| plugins |
| project_branches |
+-----+
```

Ask Gemini

54.237.135.114:8081/manage/pluginManager/updates/

Jenkins

Manage Jenkins

Plugins

Plugins

Download progress

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Preparation

Apache HttpComponents Client 5.x API

Slack Notification

Loading plugin extensions

• Checking internet connectivity

• Checking update center connectivity

• Success

Success

Success

Success

→ [Go back to the top page](#)

(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

REST APIJenkins 2.568.3

30°

10:5003-10-2026

api.slack.com/apps

slack api

Your Apps

DocumentationTutorialsYour Apps

Create AI agent

Help desk

Starter agent

Agent name

jenkins-bot

Letters, spaces, numbers, dashes, and parentheses only. You can change this later.

Workspace

jenkins-bot

This can't be changed later. If you leave this workspace, you'll lose access to this app.

Step 1 of 2

Back

Review Agent

31°

11:2103-10-2026

Workspace ?

jenkins-bot

Credential ?

jenkins-bot-token

Default channel / member id ?

all-jenkins-bot

☒ Custom slack app bot user ?

Advanced

Success

Test Connection

Save Apply

Jenkins / hiring\_app / #9

```
INFO: Analysis report compressed in 37ms, zip size=34.6 kB
INFO: Analysis report uploaded in 29ms
INFO: ANALYSIS SUCCESSFUL, you can find the results at: http://34.235.166.178:9002/dashboard?id=Test
INFO: Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
INFO: More about the report processing at http://34.235.166.178:9002/api/ce/task?id=717d2cb3-539a-4239-9023-f674f8503c6a
INFO: Analysis total time: 5.850 s
INFO: -----
INFO: EXECUTION SUCCESS
INFO: -----
INFO: Total time: 8.184s
INFO: Final Memory: 17M/60M
INFO: -----
[Slack Notifications] found #8 as previous completed, non-aborted build
[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it
Finished: SUCCESS
```

Jenkins 2.568.3

```
[ec2-user@ip-172-31-16-190 ~]$ cd /var/lib/jenkins/
[ec2-user@ip-172-31-16-190 jenkins]$ cd workspace/
[ec2-user@ip-172-31-16-190 workspace]$ ls
first_job hiring_app v-profile
[ec2-user@ip-172-31-16-190 workspace]$ cd hiring_app/
[ec2-user@ip-172-31-16-190 hiring_app]$ ls
Dockerfile Jenkinsfile README.md 'Untitled Diagram.drawio' jenkinsfile-cicd pom.xml src
[ec2-user@ip-172-31-16-190 hiring_app]$
```

